HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Exercise 29.08

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Exercise 1

Given statements:

(1) Document 1 was created by Minh.
(2) Document 2 and Document 3 were created by the same author, whose identity is unknown.
(3) Document 3 states that Document 1 was published by the W3C.

The three documents are identified by the URIs doc1, doc2 and doc3. Authorship and publication are described with the Dublin Core terms dcterms:creator and dcterms:publisher.

Namespaces used

Prefix	Namespace URI
`rdf:`	`http://www.w3.org/1999/02/22-rdf-syntax-ns#`
`dcterms:`	`http://purl.org/dc/terms/`
`ex:`	`http://example.org/`

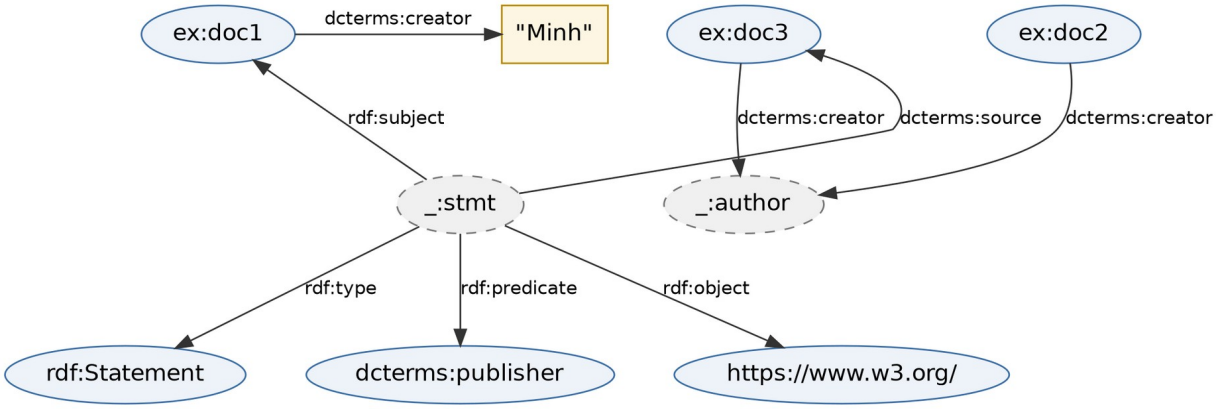
Modelling remarks

Statement (2) mentions an author who exists but is not identified. In RDF this is exactly what a blank node is for: doc2 and doc3 both point to the same anonymous node _:author, which records that the two documents share one author without naming that author.

Statement (3) is not a statement about Document 1; it is a statement about a statement. The graph must record that doc3 asserts "doc1 was published by the W3C", not that doc1 was in fact published by the W3C. Writing the triple ex:doc1 dcterms:publisher <https://www.w3.org/> directly would commit the graph to a fact that the exercise does not give. The standard RDF solution is reification: an instance of rdf:Statement describes the triple through rdf:subject, rdf:predicate and rdf:object, and dcterms:source links it back to doc3 as the document that reports it.

Task 1. RDF graph

Ellipses are resources, the rectangle is a literal, and dashed ellipses are blank nodes. Arrows are directed from subject to object and are labelled with the predicate.



The graph contains the following eight triples:

Subject	Predicate	Object
ex:doc1	dcterms:creator	"Minh"
ex:doc2	dcterms:creator	_:author
ex:doc3	dcterms:creator	_:author
_:stmt	rdf:type	rdf:Statement
_:stmt	rdf:subject	ex:doc1
_:stmt	rdf:predicate	dcterms:publisher
_:stmt	rdf:object	https://www.w3.org/
_:stmt	dcterms:source	ex:doc3

Task 2. Turtle notation

```
@prefix rdf:     <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix dcterms: <http://purl.org/dc/terms/> .
@prefix ex:      <http://example.org/> .

# Statement 1: Document 1 was created by Minh.
ex:doc1 dcterms:creator "Minh" .

# Statement 2: Document 2 and Document 3 share the same author,
# whose identity is unknown -> a blank node is used.
ex:doc2 dcterms:creator _:author .
ex:doc3 dcterms:creator _:author .

# Statement 3: Document 3 states that Document 1 was published by the W3C.
# The publication fact is only reported by doc3, so it is reified
# instead of being asserted directly in the graph.
_:stmt a rdf:Statement ;
       rdf:subject    ex:doc1 ;
       rdf:predicate  dcterms:publisher ;
       rdf:object     <https://www.w3.org/> ;
       dcterms:source ex:doc3 .
```

Task 3. RDF/XML notation

```
<?xml version="1.0" encoding="UTF-8"?>
<rdf:RDF
    xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
    xmlns:dcterms="http://purl.org/dc/terms/"
    xmlns:ex="http://example.org/">

  <!-- Document 1 was created by Minh -->
  <rdf:Description rdf:about="http://example.org/doc1">
    <dcterms:creator>Minh</dcterms:creator>
  </rdf:Description>

  <!-- Document 2 and Document 3 share one unknown author -->
  <rdf:Description rdf:about="http://example.org/doc2">
    <dcterms:creator rdf:nodeID="author"/>
  </rdf:Description>

  <rdf:Description rdf:about="http://example.org/doc3">
    <dcterms:creator rdf:nodeID="author"/>
  </rdf:Description>

  <!-- Document 3 states that Document 1 was published by the W3C -->
  <rdf:Statement rdf:nodeID="stmt">
    <rdf:subject rdf:resource="http://example.org/doc1"/>
    <rdf:predicate rdf:resource="http://purl.org/dc/terms/publisher"/>
    <rdf:object rdf:resource="https://www.w3.org/"/>
    <dcterms:source rdf:resource="http://example.org/doc3"/>
  </rdf:Statement>

</rdf:RDF>
```

Note on dcterms:creator

In the DCMI terms vocabulary, the range of dcterms:creator is dcterms:Agent, so a strict modelling would give Minh a resource of her own rather than a plain literal, for example:

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

ex:doc1 dcterms:creator [ a dcterms:Agent ;
                          rdfs:label "Minh" ] .
```

The literal form used above follows the legacy Dublin Core Element Set (http://purl.org/dc/elements/1.1/), where a literal value is allowed. Both forms are accepted in practice; the literal form is kept here because it stays closer to the wording of the exercise.